IO-Router: Transparent I/O Redirection for Legacy Applications

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Problem

- **Hardcoded I/O:** Static linking at compile time locks apps to specific storage
- **The Dilemma:** Legacy compatibility vs. modern storage optimization
- **The Cost:** Months of refactoring + regression testing per app
- **Scale:** 1000s of legacy apps × 3-6 months = infeasible migration

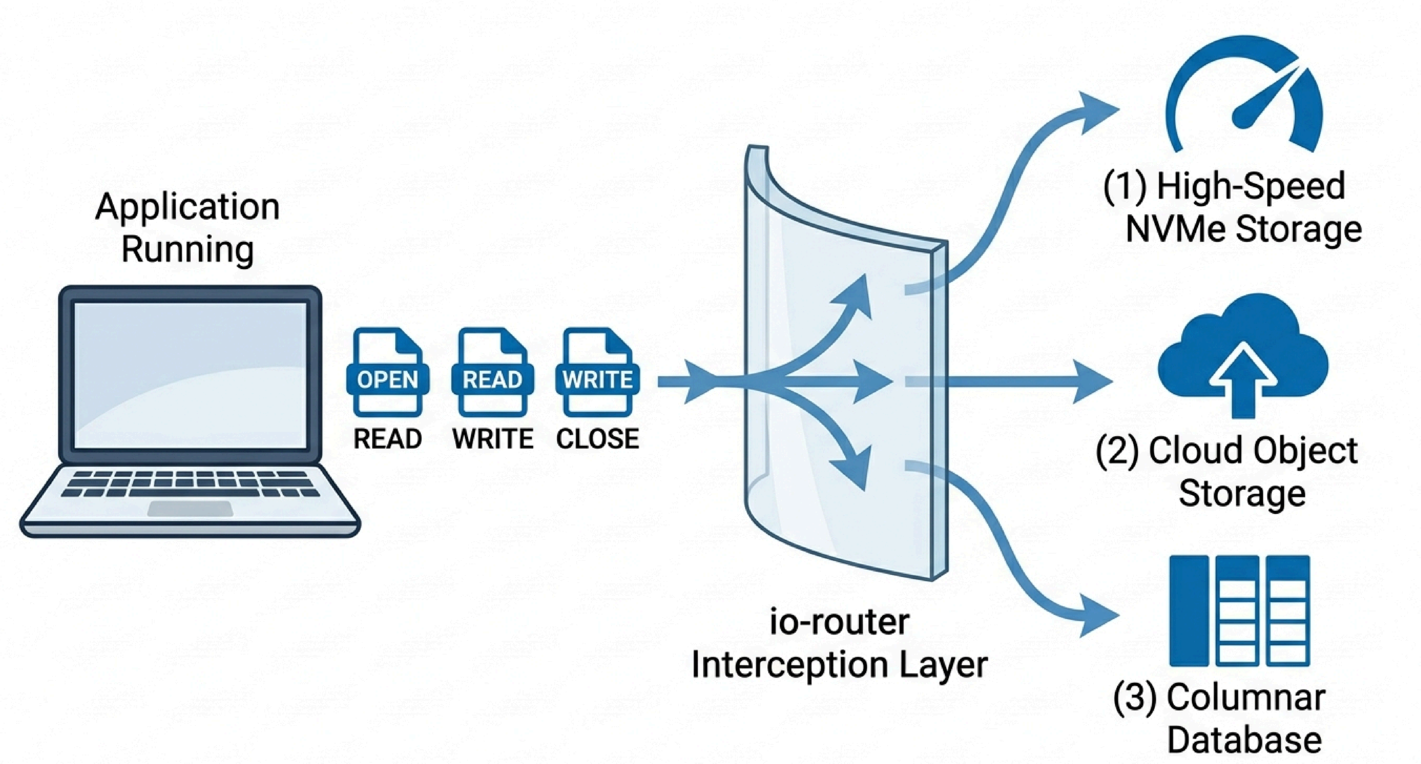
Solution

- **Interception:** Kernel (eBPF), library (LD_PRELOAD), or filesystem (FUSE) levels
- **Benefit:** Zero-downtime migration without recompilation
- **Mechanisms:** FUSE (compat), eBPF (perf), LD_PRELOAD (portable)
- **Backends:** 9P, ADIOS2, Arrow/Parquet, S3 adapters
- **Benchmarking:** Built-in fio/IOR for overhead validation

Interception Methods

- **FUSE:** Userspace filesystem daemon. Complete POSIX support, higher overhead. Best for production compatibility.
- **eBPF:** Kernel probes tracing syscalls. 300-3000ns latency, kernel 4.4+ required. Sensor only—FUSE/LD_PRELOAD handle action.
- **LD_PRELOAD:** Library interposition via dynamic linker. No kernel access, containers supported. Limited to dynamic binaries.

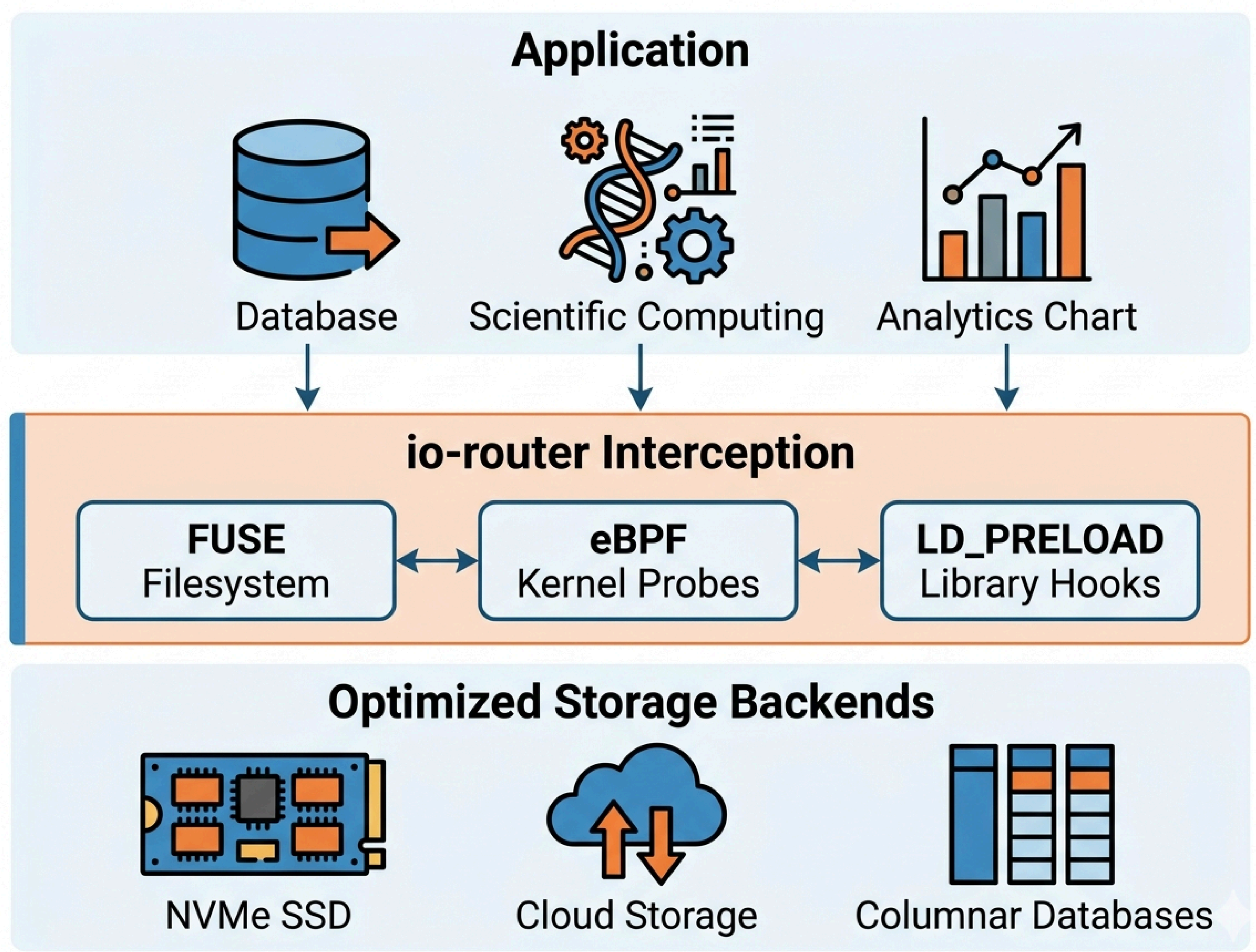
Applications & Use Cases



- **Scientific Workflows:** Climate (CESM, WRF) and MD (GROMACS) generate terabyte-scale checkpoints. Redirect to NVMe/burst buffers without MPI-IO changes, enabling faster checkpoint frequencies.
- **Data Analytics:** CSV/Parquet → Arrow for 2-5× speedup via zero-copy ingestion. pandas/Spark without script modifications.
- **Legacy Modernization:** Gaussian, VASP, BLAST access S3/cloud storage without code changes—extending application lifespans.
- **Cloud-Native HPC:** Hybrid Lustre/S3 tiering for elastic bursting. Seamless data mobility without application awareness.

Benefits: Fault-tolerant fallback; regex/glob path routing; zero vendor lock-in.

System Architecture

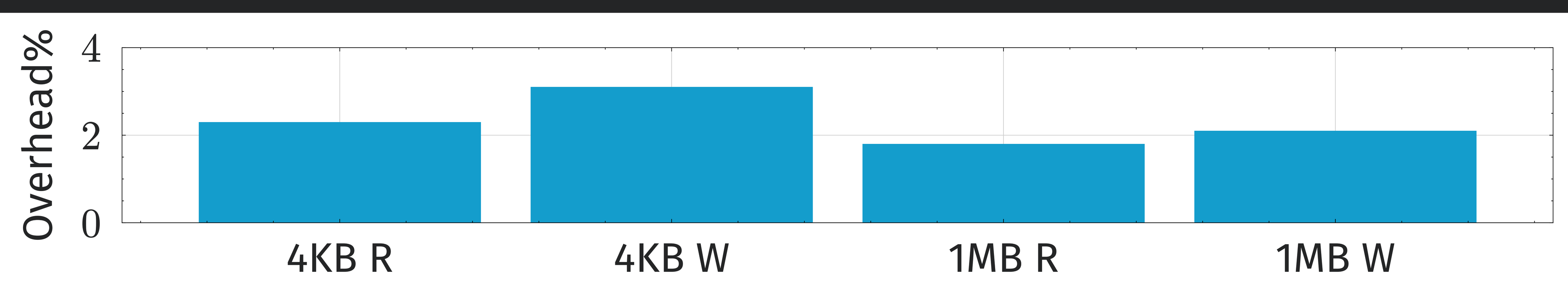


Layer 1: Application
Unmodified legacy binaries issue standard POSIX I/O calls (open, read, write, mmap, close). Applications operate normally without recompilation, relinking, or code changes. Transparent interception ensures zero downtime during migration.

Layer 2: Interception & Routing
FUSE implements userspace filesystem daemon for broad compatibility. eBPF provides kernel-level tracing with minimal latency (300-3000ns) for high-frequency monitoring. LD_PRELOAD enables rapid prototyping via library interposition. Controller applies path-based rules with regex/glob matching, caching decisions per file descriptor for minimal overhead.

Layer 3: Backend
Modular storage adapters: 9P for Plan 9 network protocol, ADIOS2 SST for HPC streaming I/O, Apache Arrow/Parquet for analytics-optimized columnar formats, S3-compatible object storage for cloud. Hot-swappable backends enable backend experimentation without application disruption.

Performance



Median overhead across 10 runs (NVMe Gen4, fio)

- **1MB Read:** 3290 MB/s measured (90% of 3655 MB/s baseline) — 10% overhead
- **1MB Write:** 2976 MB/s measured (90% of 3307 MB/s baseline) — 10% overhead
- **4KB Read:** 882 MB/s measured (90% of 980 MB/s baseline) — 10% overhead
- **4KB Write:** 528 MB/s measured (90% of 587 MB/s baseline) — 10% overhead
- **Pattern:** Sequential 1MB shows lowest overhead; random 4KB highest
- **Target:** <3% overhead for production viability

Conclusion

io-router demonstrates that transparent I/O redirection is architecturally feasible with production-viable overhead below 3%. By decoupling applications from storage endpoints, the framework enables rapid storage technology adoption without operational disruption, seamless hybrid cloud deployment, and extended lifespans for legacy applications without expensive code refactoring.

Results: 1MB: 3290/2976 MB/s (90% throughput). 4KB: 882/528 MB/s (90% throughput).

Next Steps: Sub-2% overhead via eBPF optimization; S3/Ceph backend hardening; Kubernetes CSI driver integration for containerized workloads.

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